

Markov Chains

Geometric Algorithms

Practice Problem

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + x_2 \\ 2x_2 \\ x_2 + bx_3 \end{bmatrix}$$

For what values of b is the above transformation singular? Explain your answer

non-invertible

Find the inverse of the matrix implementing the above transformation, given $b = 1$

Solution

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & b \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 1 & 0 \\ 0 & \boxed{2} & 0 \\ 0 & 0 & b \end{bmatrix}$$

$$b = 0$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + x_2 \\ 2x_2 \\ x_2 + bx_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 & | & 1 & 0 & 0 \\ 0 & 2 & 0 & | & 0 & 1 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & | & 1 & 0 & 0 \\ 0 & 1 & 0 & | & 0 & 1/2 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & | & 1 & 1/2 & 0 \\ 0 & 1 & 0 & | & 0 & 1/2 & 0 \\ 0 & 0 & 1 & | & 0 & 0 & 1 \end{bmatrix}$$

Objectives

Finish up our discussion of LU factorization

Motivate linear dynamical systems

Analyze Markov chains and their properties

Learn to solve for steady-states of Markov chains

Keywords

linear dynamical systems

recurrence relations

linear difference equations

state vector

probability vector

stochastic matrix

Markov chain

steady-state vector

random walk

state diagram

Analyzing LU Factorization

Recall: LU Factorization

Given a $m \times n$ matrix A , we are going to factorize A as

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{\substack{\text{echelon form of } A \\ U}}$$

Note. This applies to non-square matrices

Analyzing Linear Algebra Algorithms

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We will not use $O(\cdot)$ notation!

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Operations (FLOPs):

- » addition
- » subtraction
- » multiplication
- » division
- » square root

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$2n$ vs. n is very different
when $n \sim 10^{20}$

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for polynomials, they are equivalent to their dominant term

Dominant Terms

the dominant term of a polynomial is the monomial with the highest degree

$$\lim_{i \rightarrow \infty} \frac{3x^3 + 100000x^2}{3x^3} = 1$$

$3x^3$ dominates the function even though the coefficient for x^2 is so large

How To: Solving systems with the LU

Question. Solve the equation $A\mathbf{x} = \mathbf{b}$ given that $A = LU$ is a LU factorization

Solution. First solve $L\mathbf{x} = \mathbf{b}$ to get a solution $\mathbf{c}$, then solve $U\mathbf{x} = \mathbf{c}$ to get a solution $\mathbf{d}$

$$L\vec{x} = \vec{b}$$

$$L\vec{c} = \vec{b}$$

$$A\vec{d} = \vec{b}$$

$$U\vec{x} = \vec{c}$$

$$L\vec{d} = \vec{c}$$

$$LU\vec{d} = L\vec{c} = \vec{b}$$

FLOPs for Solving General Systems

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Gaussian Elimination: $\sim \frac{2n^3}{3}$ FLOPS

The following FLOP
estimates are based
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$$A \tilde{x} = b \quad \tilde{x} = A^{-1} b$$

Solving by matrix inversion: $\sim 2n^3$ FLOPS

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Solving by LU Factorization: $\sim \frac{2n^3}{3}$ FLOPS

If you solve several matrix equations for the same matrix, LU factorization is faster than matrix inversion on the first equation, and the same (asymptotically) in later equations (and it works for rectangular matrices)

Don't actually invert matrices

Another Consideration: Density

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If A is **sparse** then L and U will be too

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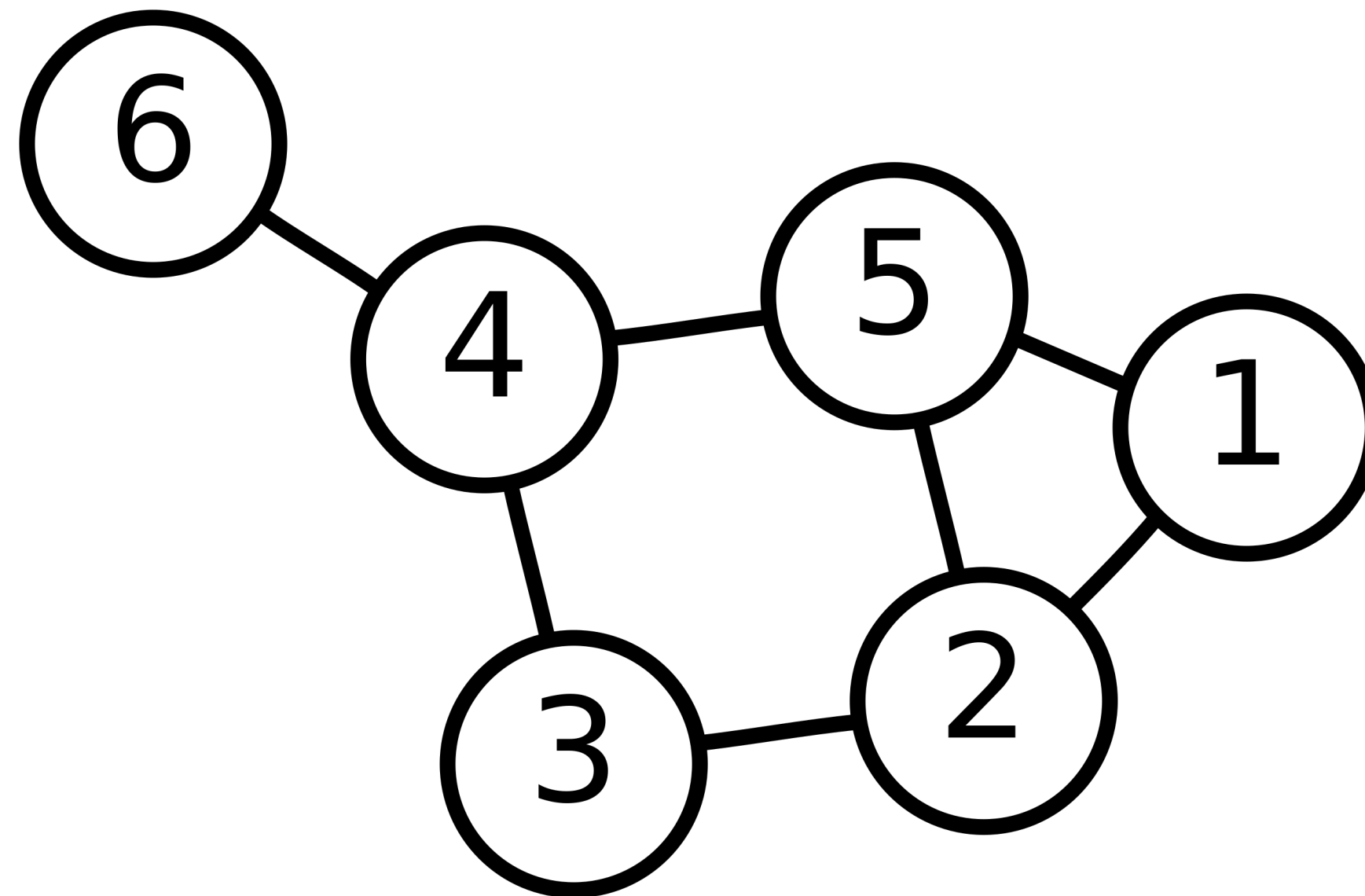
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(This is the topic of Lab 3)

Graph Theory Primer

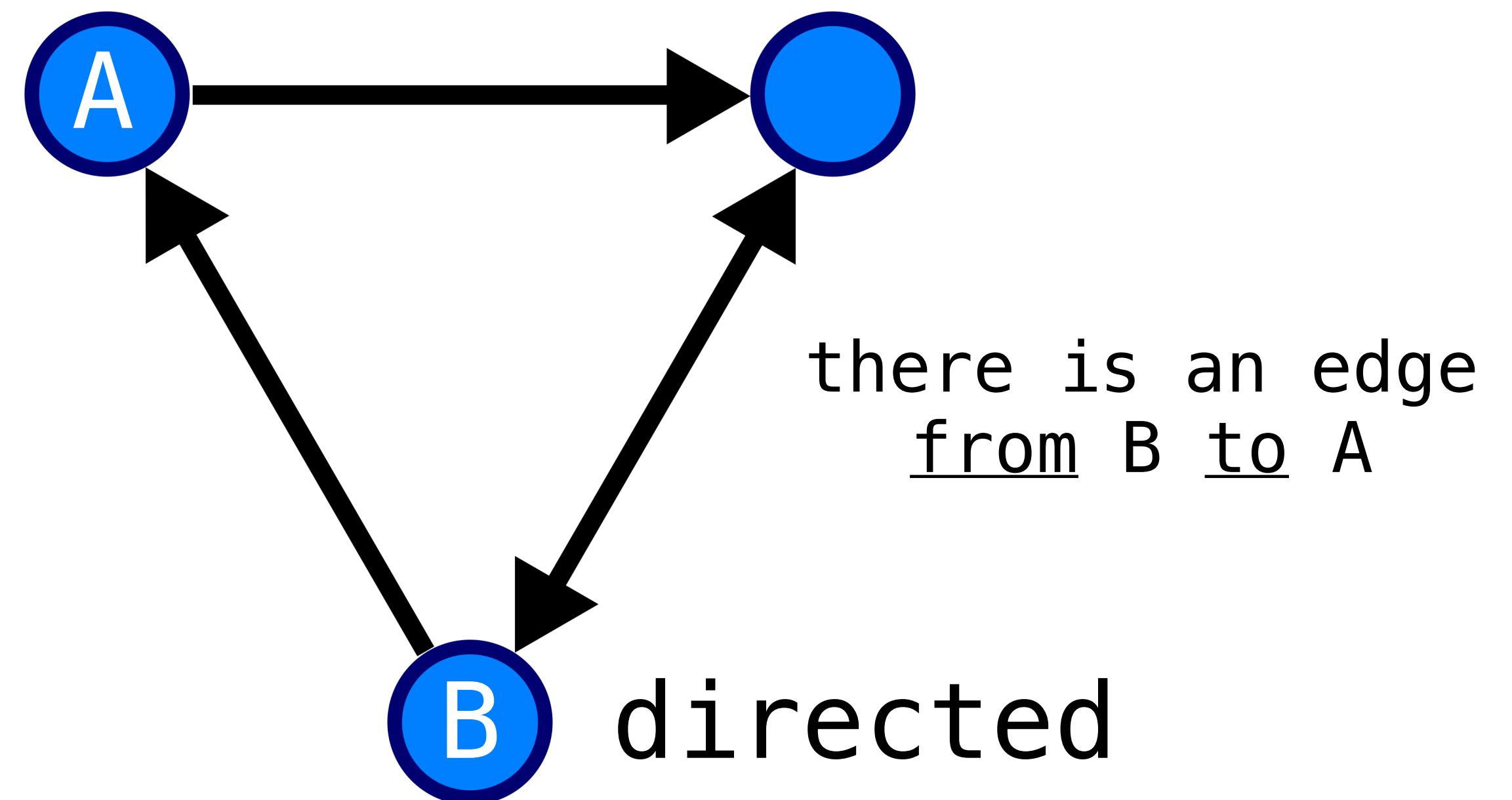
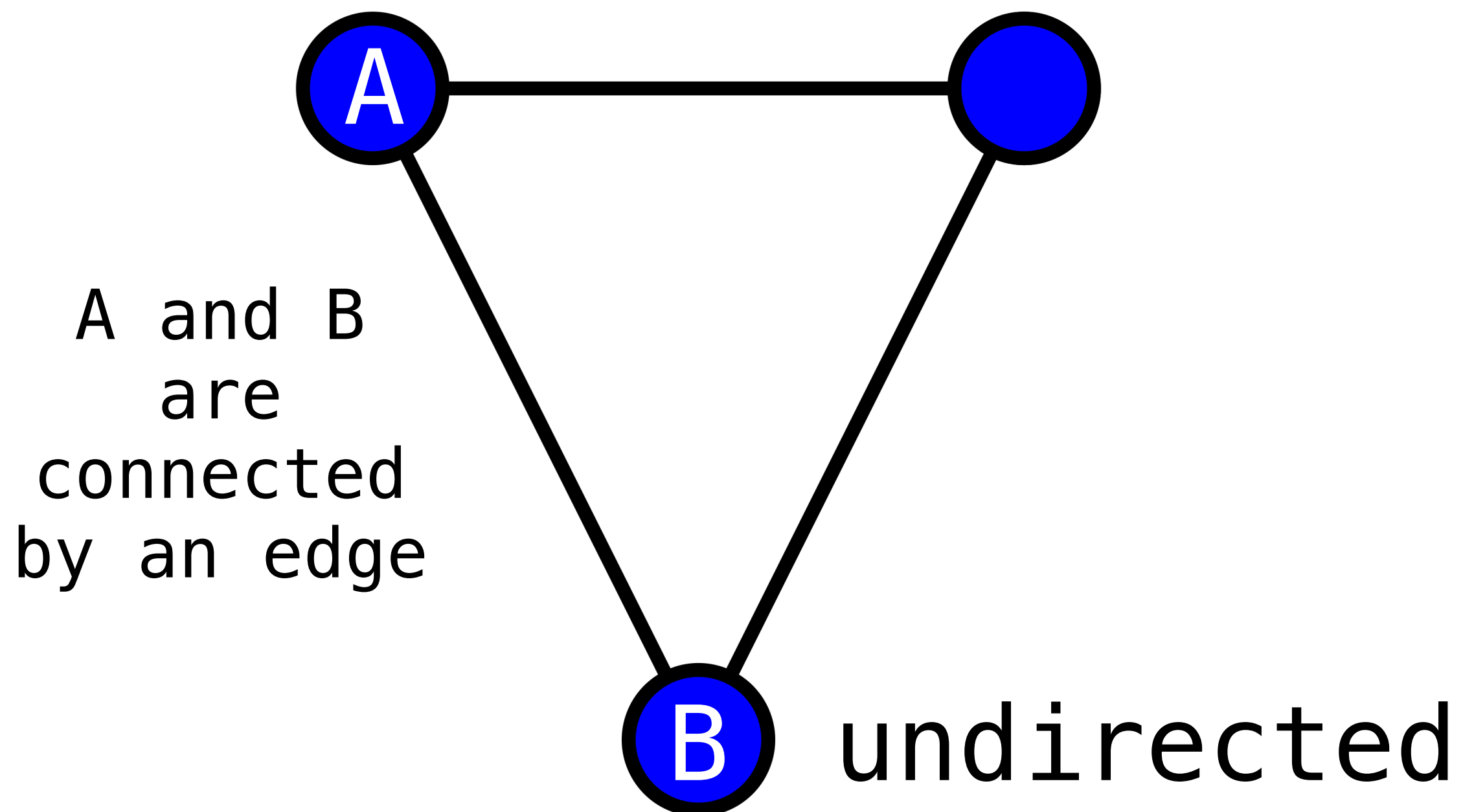
Graphs

Def (Informal). A graph is a collection of nodes with edges between them



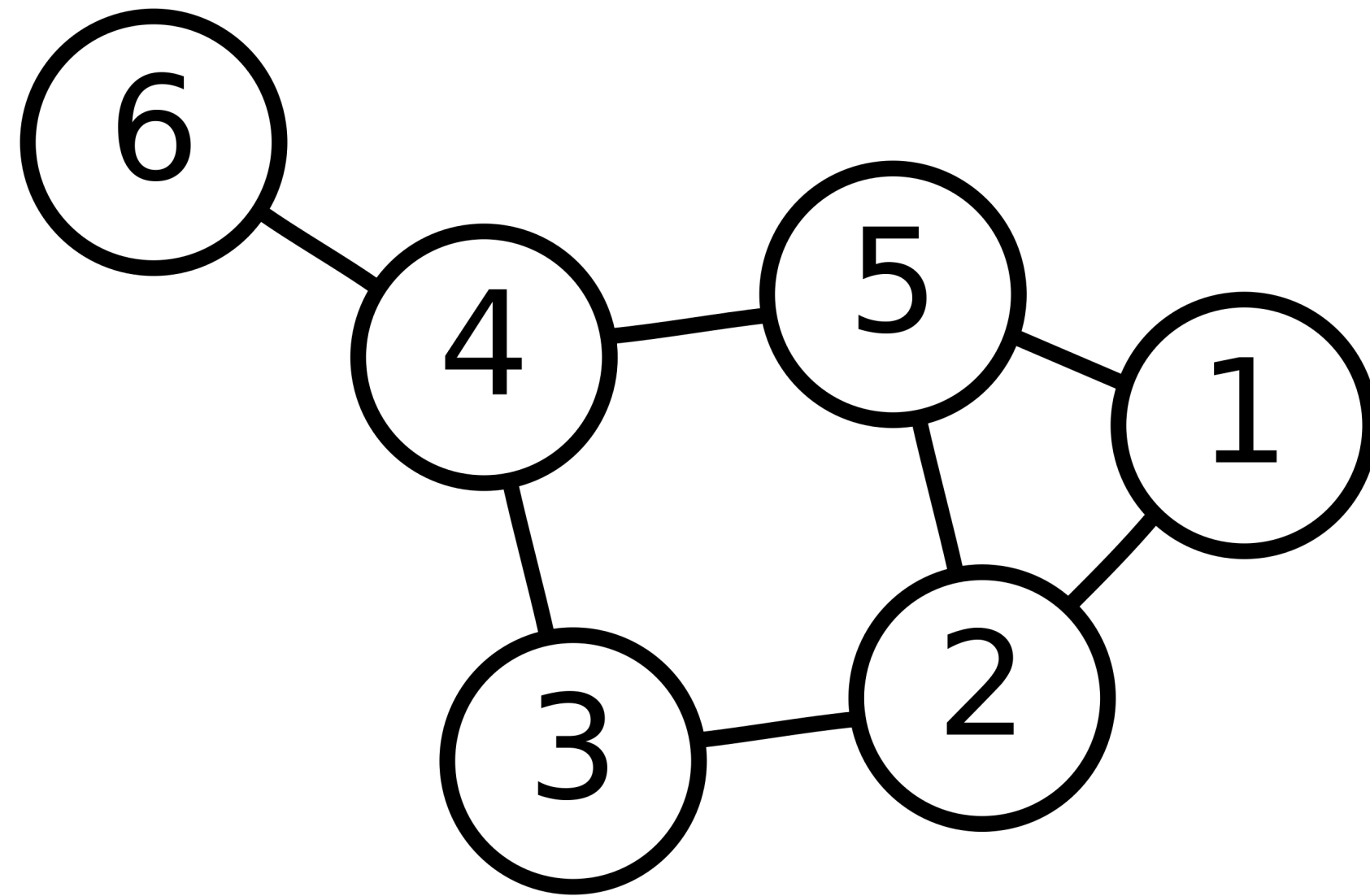
Directed vs. Undirected Graphs

A graph is **directed** if its edges have a direction

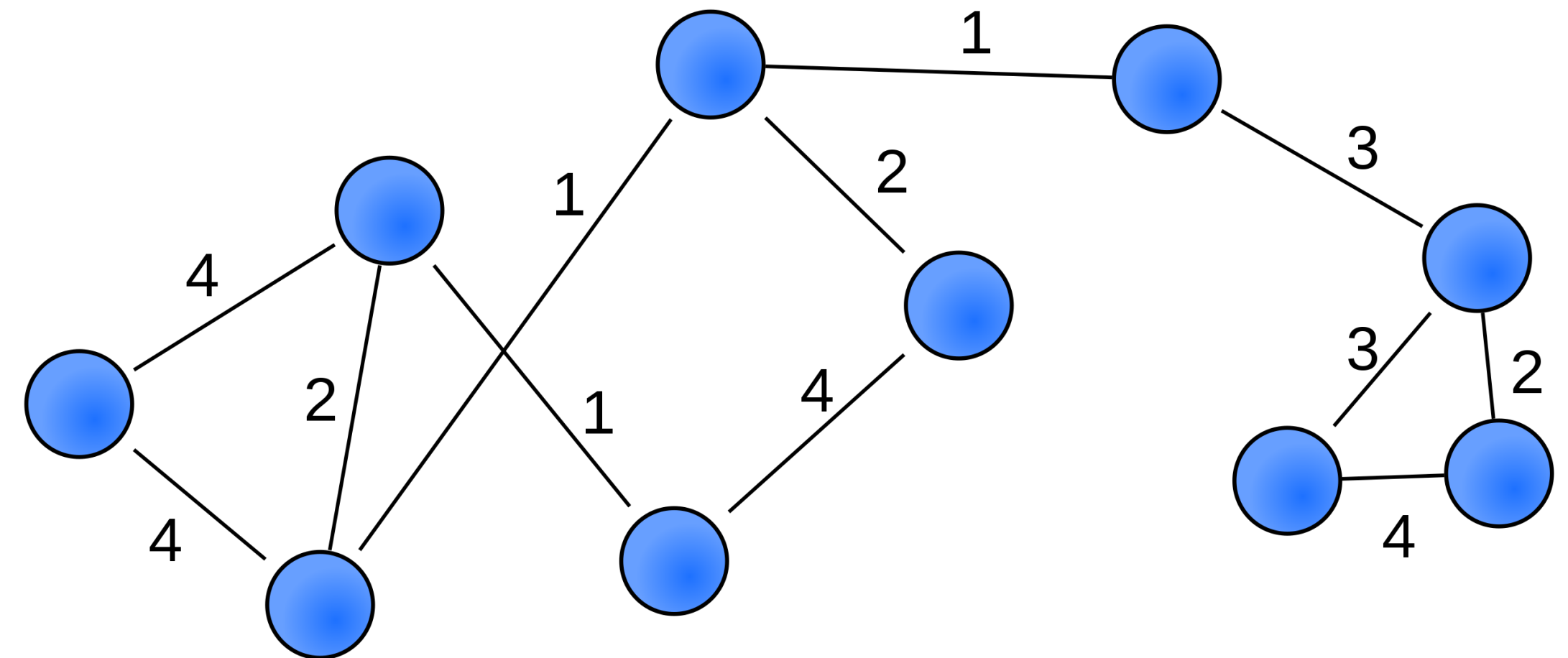


Weighted vs Unweighted graphs

A graph is **weighted** if its edges have associated values



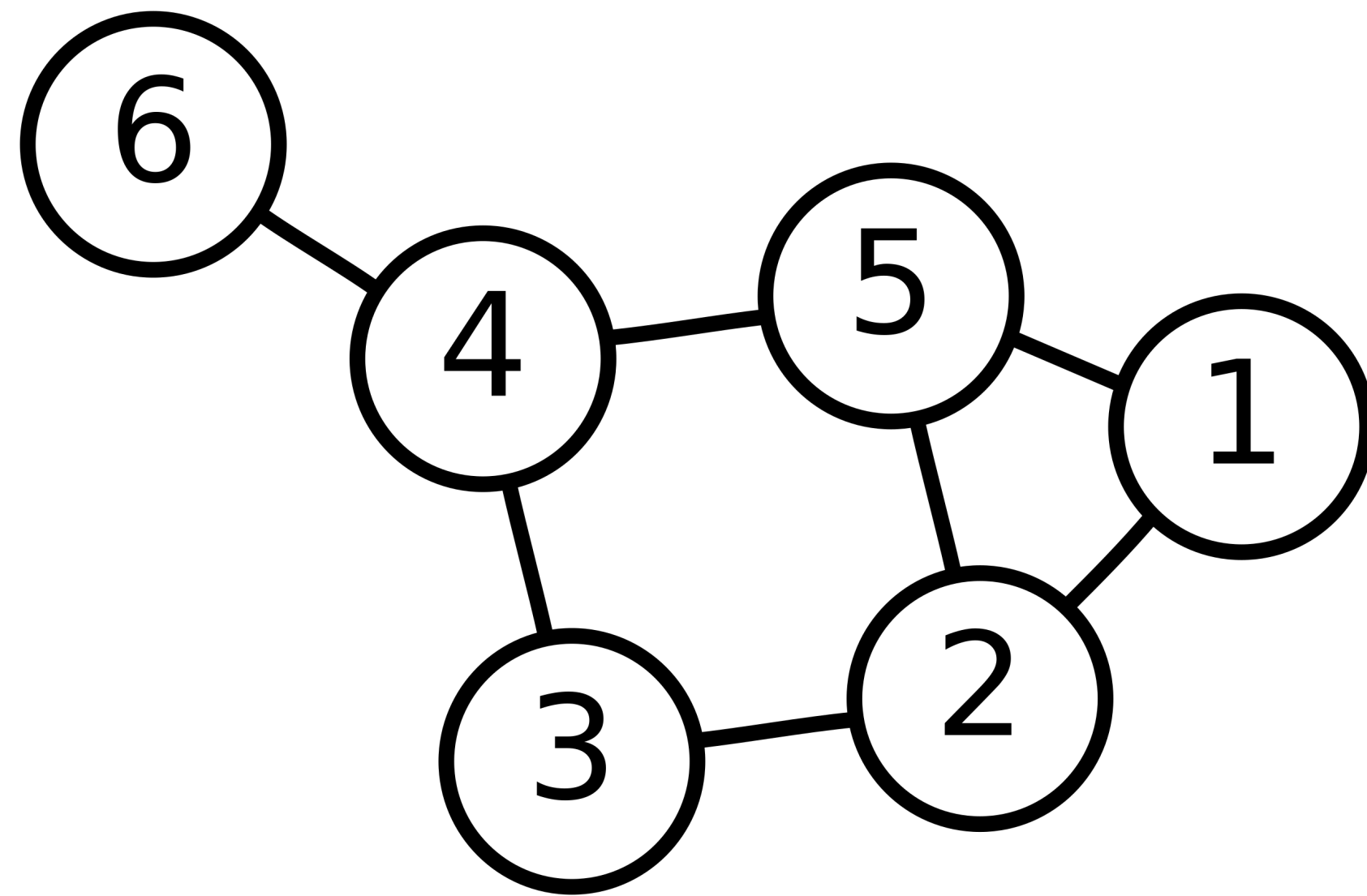
unweighted



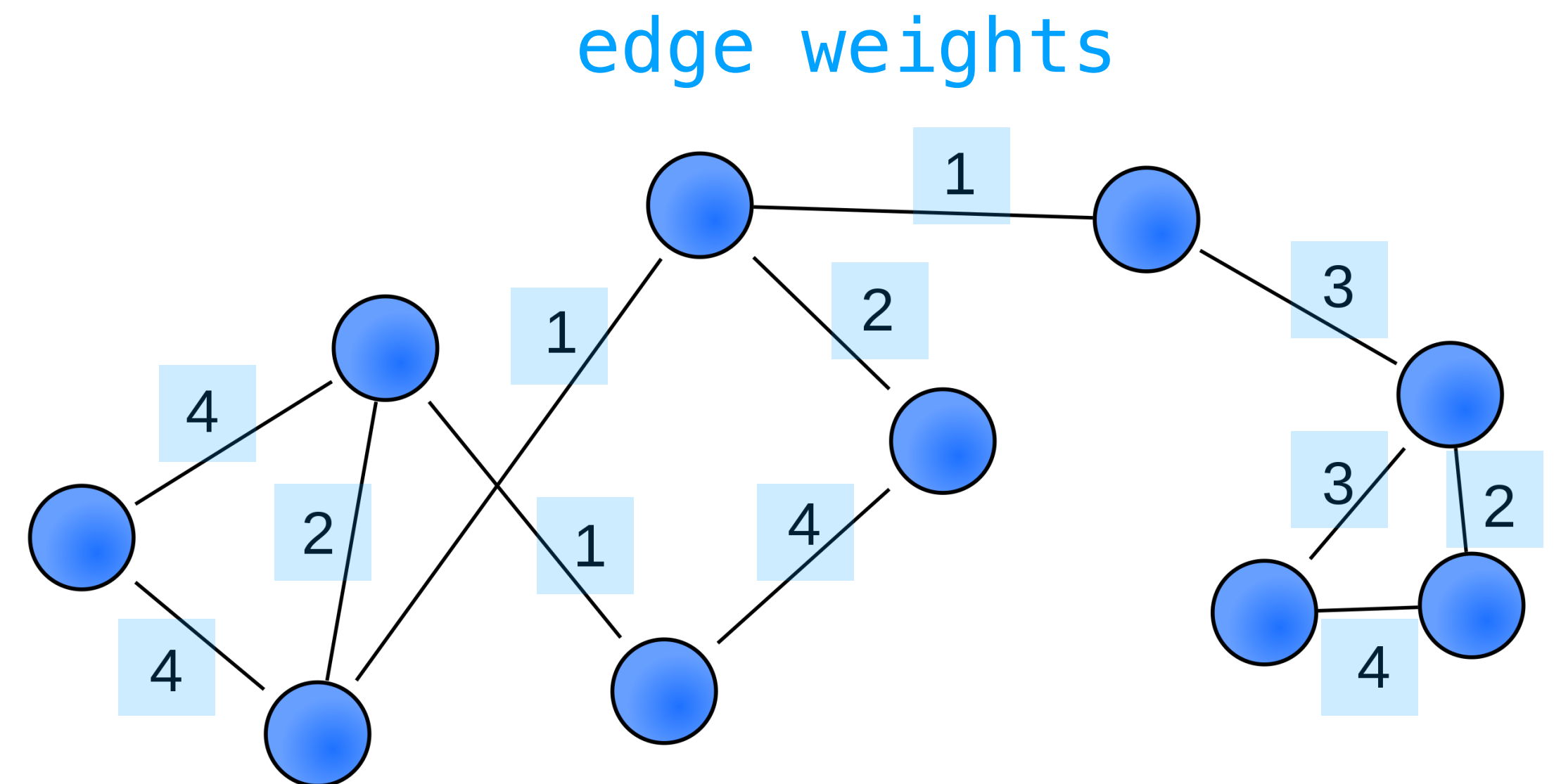
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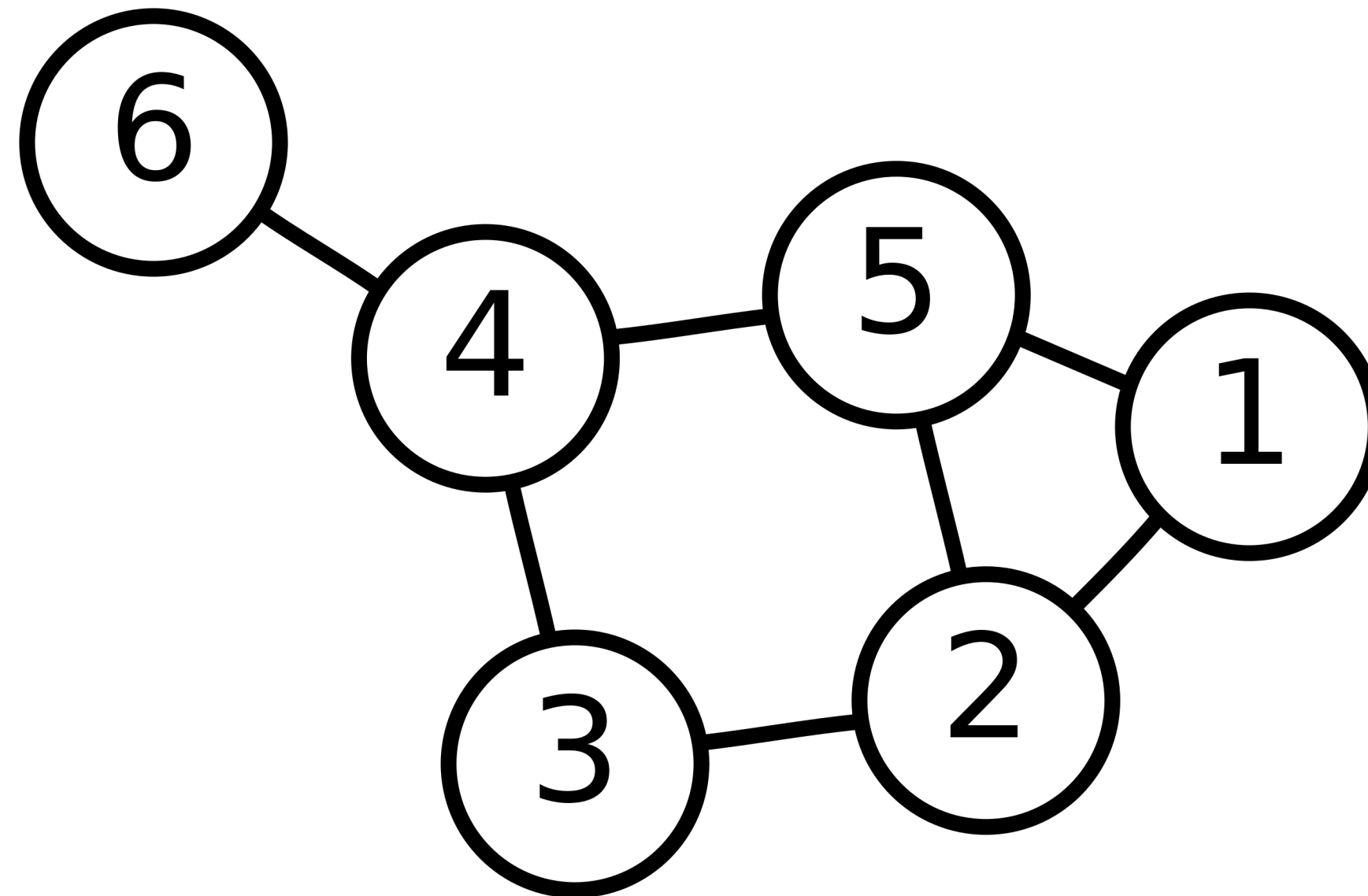
unweighted



weighted

Simple Graphs

A graph is **simple** if it is undirected, has no self loops, and no multi-edges



Four Kinds of Graphs

directed

undirected

weighted

nodes are traffic lights
edges are streets
weights are number of lanes

nodes are musicians
edges are collaborations
weights are number of collaborations

unweighted

nodes are instagram users
edges are follows

nodes are bodies of land
edges are pedestrian bridges

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Markov Chains

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Fundamental Question

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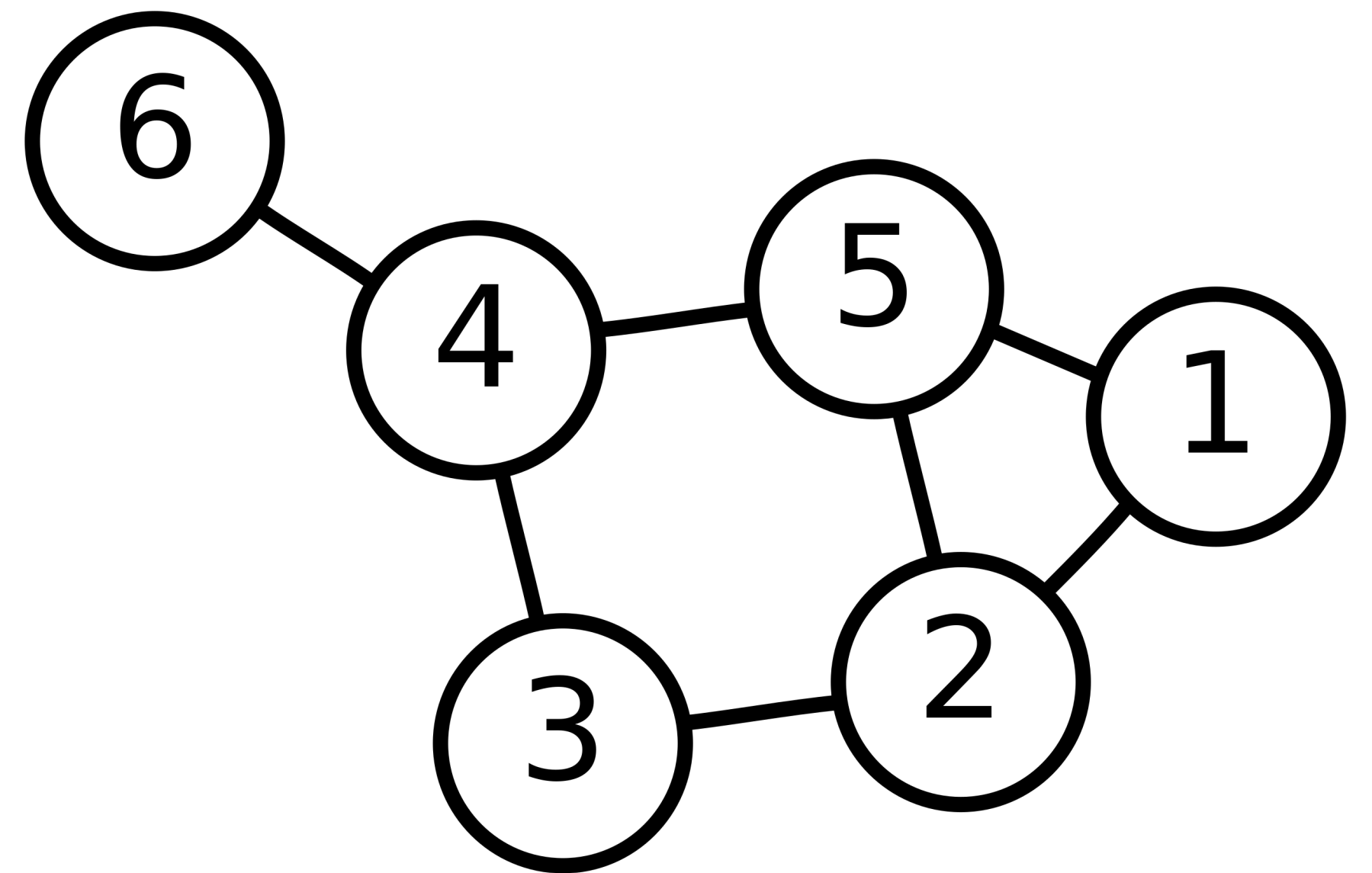
How do we represent a graph
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Fundamental Question

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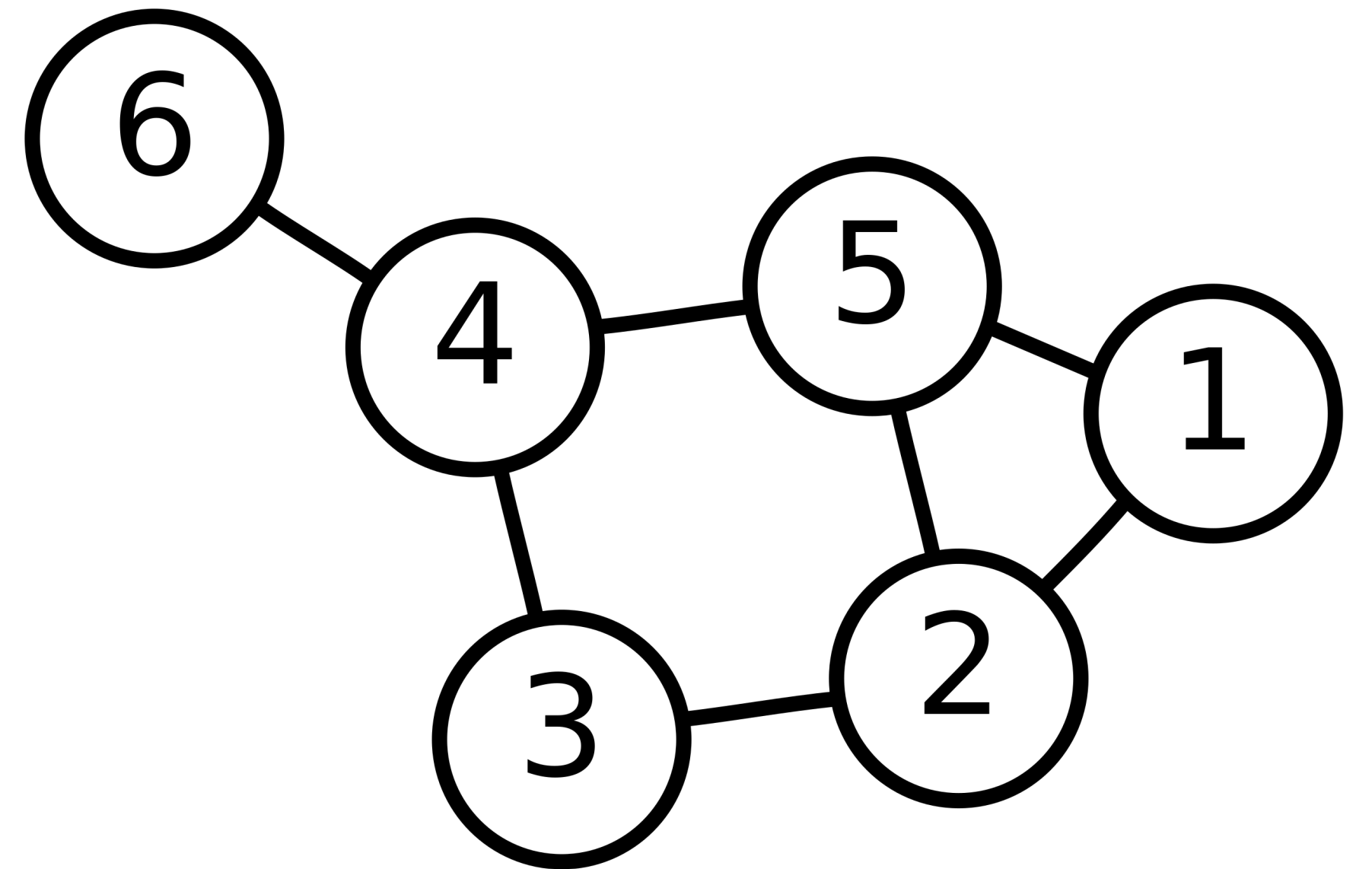
There are a couple ways, but one way is to use **matrices**

Adjacency Matrices



Adjacency Matrices

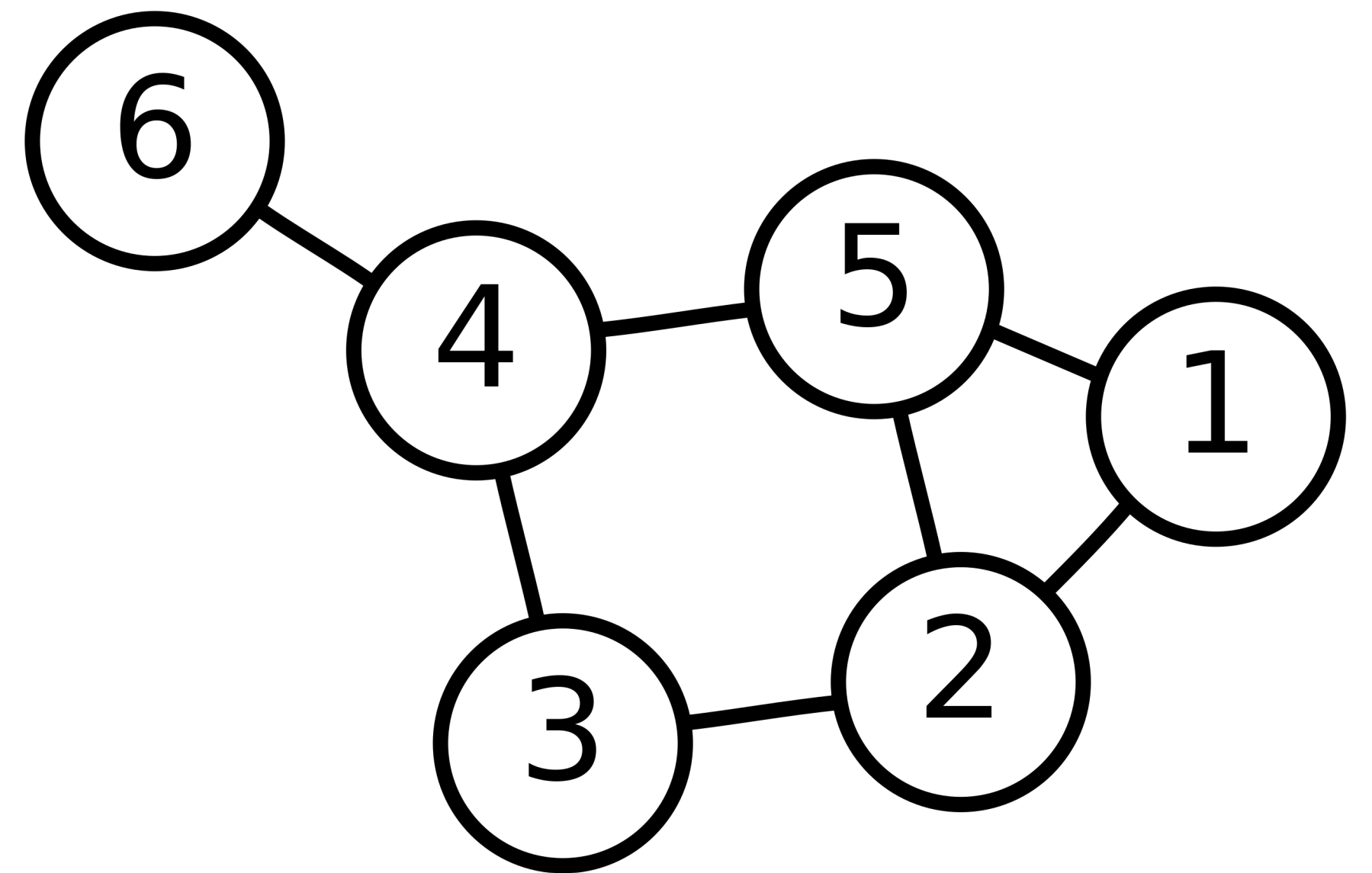
Let G be an simple graph with
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We can create the **adjacency matrix** A for G as:



Symmetrical: $A = A^T$

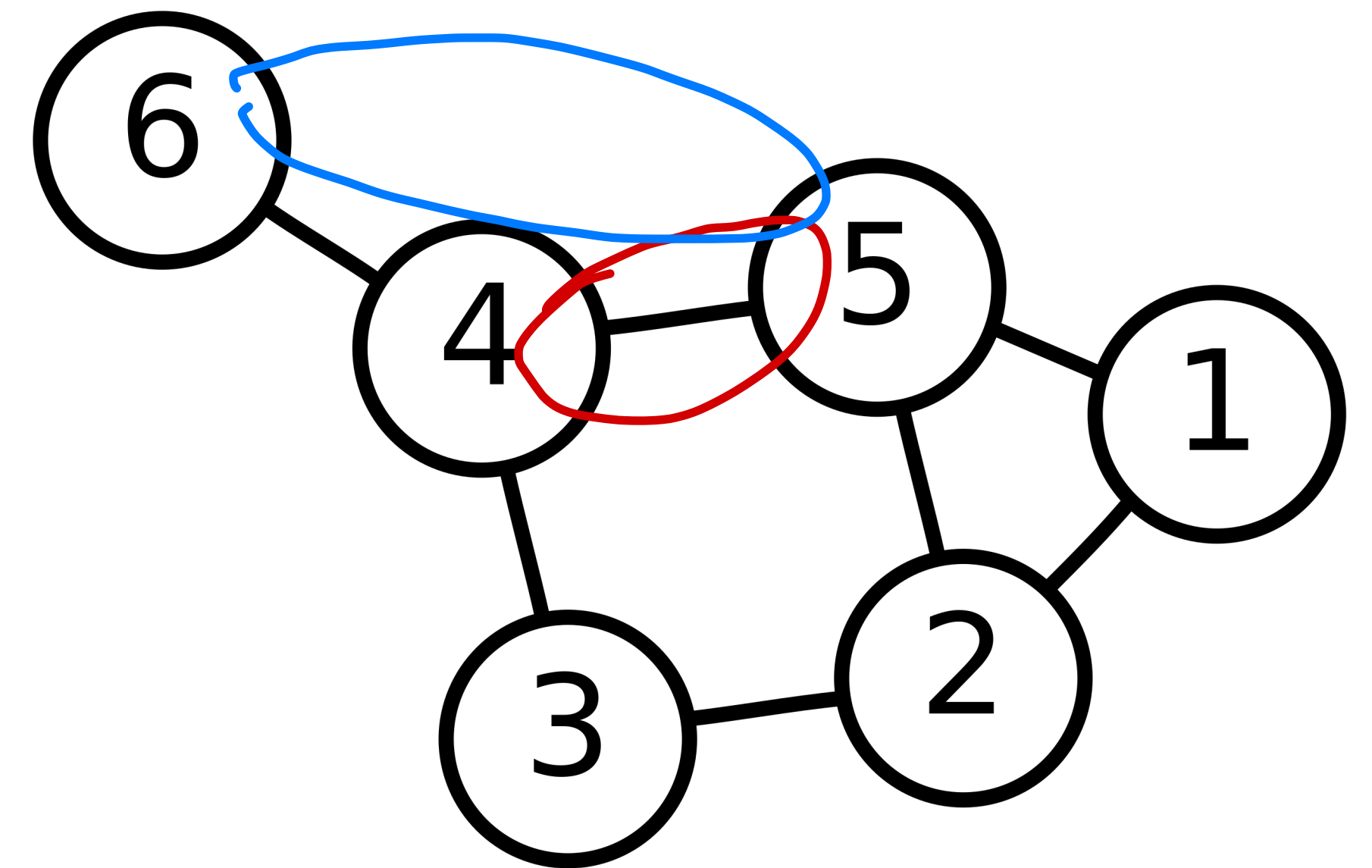
Adjacency Matrices

Let G be an simple graph with its nodes labeled by numbers 1 through n

We can create the **adjacency matrix** A for G as:

$$A_{ij} = \begin{cases} 1 & \text{edge between } i \text{ and } j \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{matrix} & & & \textcolor{red}{4} & \textcolor{blue}{5} & \textcolor{blue}{6} \\ \begin{matrix} \textcolor{red}{4} \\ \textcolor{blue}{5} \\ \textcolor{blue}{6} \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & \boxed{1} & 1 \\ 1 & 1 & 0 & \boxed{1} & 0 & \boxed{0} \\ 0 & 0 & 0 & 1 & \boxed{0} & 0 \end{bmatrix} \end{matrix}$$



Once we have adjacency matrices we
can do linear algebra on graphs

Dynamical Systems

Change

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Things change

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Things change from one state of affairs to
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Things change often in unpredictable ways

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If something changes unpredictably, what can we say about it?

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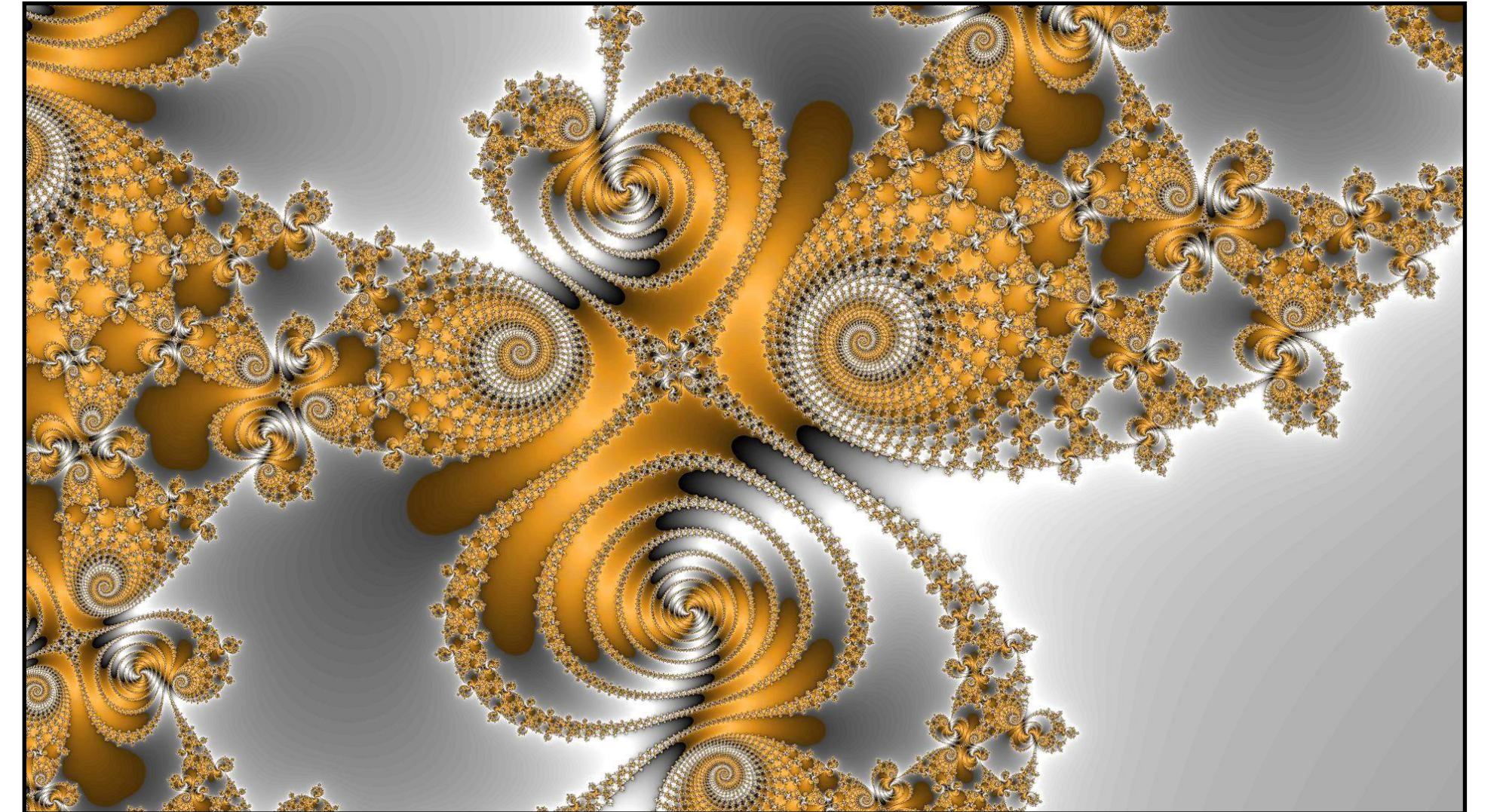
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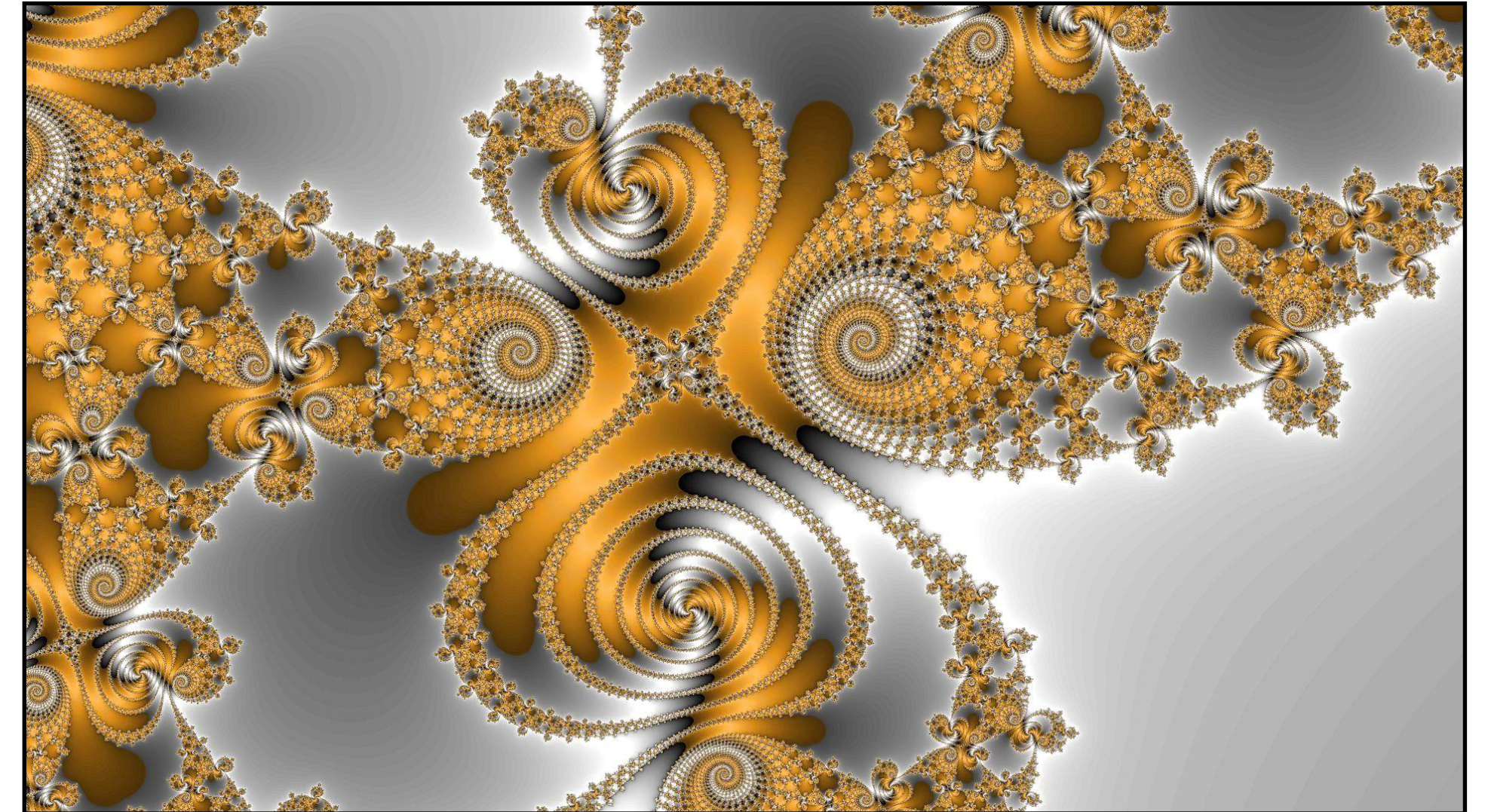
- » economics (stocks)
- » physical/chemical systems
- » populations
- » weather

An Aside: Chaos Theory



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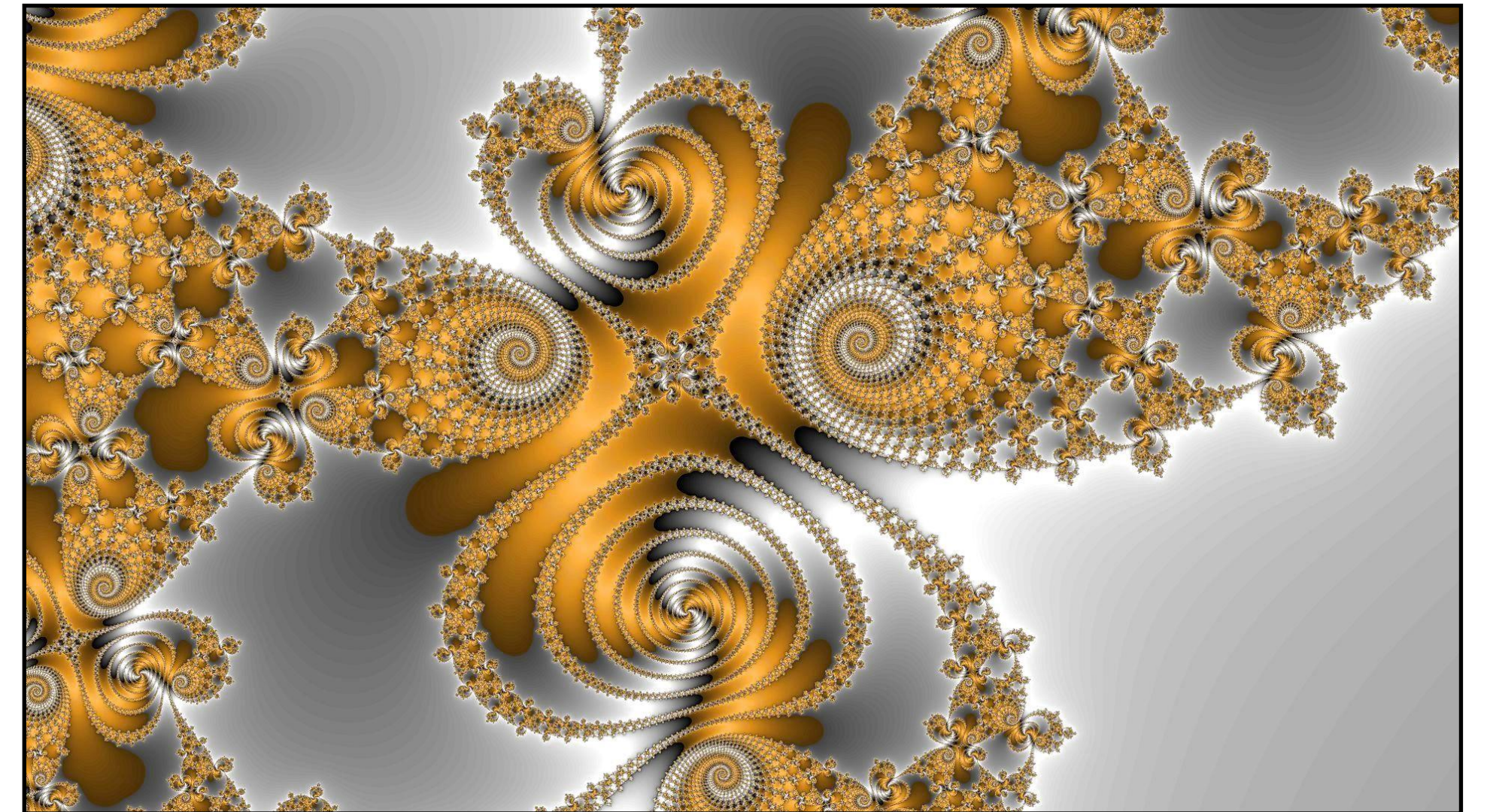
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But even in chaotic systems there are *underlying patterns* and *repeated structures*

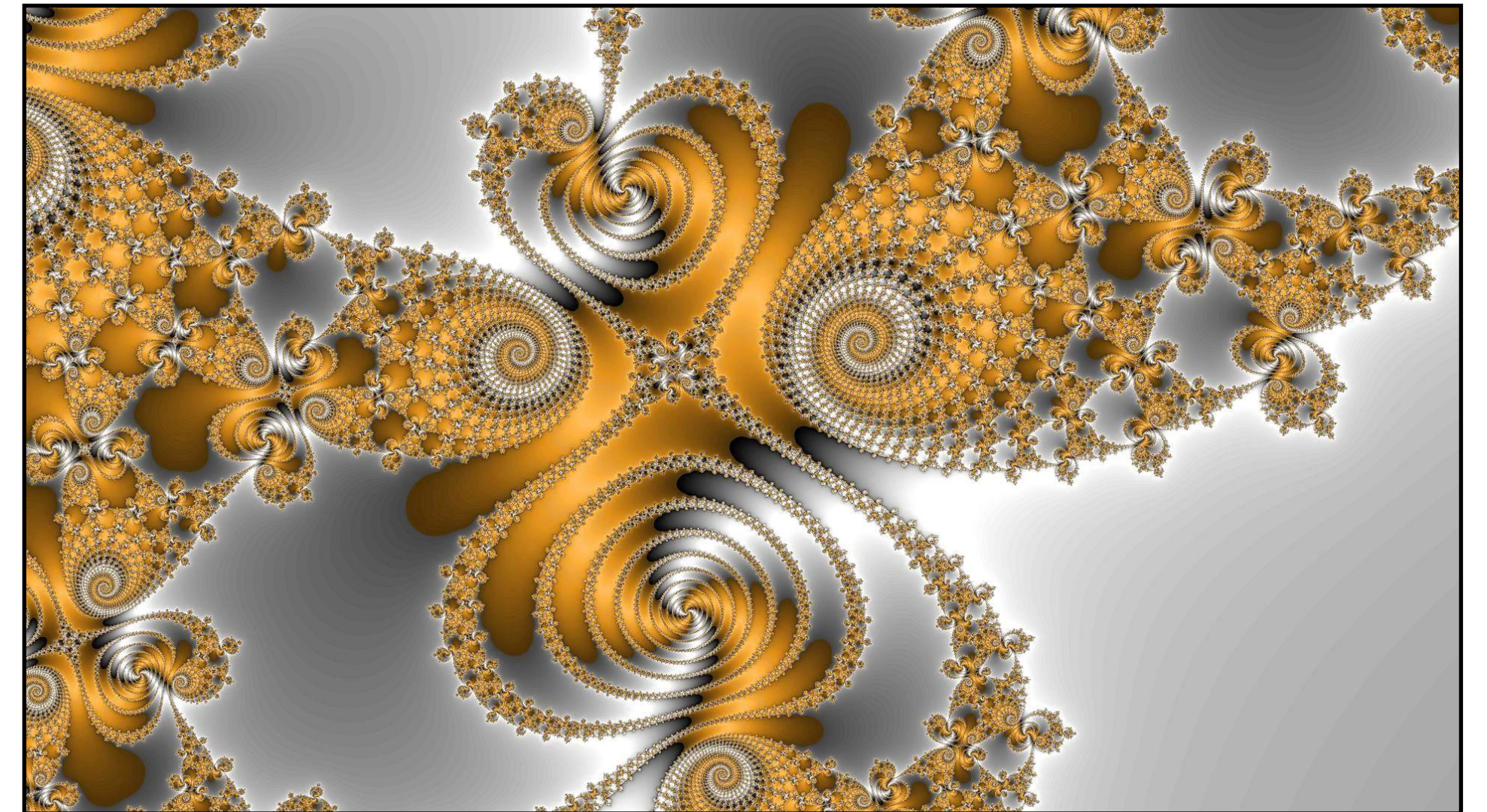


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Often it's useful to consider chaotic systems in terms of *global* properties



Motivating Questions

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Does it reach a kind of *equilibrium*?

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Does it reach a kind of *equilibrium*?

Does a part of the system *dominate* over time?

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The **possible states** of the system are vectors in $\mathbb{R}^n$

Given an **initial state vector** $\mathbf{v}_0$, we can determine the **state vector** of the system after i time steps:

$$\mathbf{v}_i = A\mathbf{v}_{i-1} \quad (\text{linear recurrence relation})$$

State Vectors

$$\mathbf{v}_0$$

$$\mathbf{v}_1 = A\mathbf{v}_0$$

$$\mathbf{v}_2 = A\mathbf{v}_1 = A(A\mathbf{v}_0)$$

$$\mathbf{v}_3 = A\mathbf{v}_2 = A(AA\mathbf{v}_0)$$

$$\mathbf{v}_4 = A\mathbf{v}_3 = A(AAA\mathbf{v}_0)$$

$$\mathbf{v}_5 = A\mathbf{v}_4 = A(AAAA\mathbf{v}_0)$$

$$\vdots$$

The **state vector** $\mathbf{v}_k$ is the state of the system after k time steps

Warm up: Population Dynamics

The Setup

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We're working for the census. We have population measurements for a city and a suburb which are geographically coincident

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We find by analyzing previous data that each year:

- » 5% of the population moves from city → suburb
- » 3% of the population moves from suburb → city

Fundamental Question

Can we make any predictions about the population of the city and suburb in 20 years?

Assumptions: No immigration, emigration, birth, death, etc. **The overall population stays fixed.**

Setting up Linear Equations

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If $\text{city}_0 = \text{city pop.} = 600,000$
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people who stayed

people who left

Setting up a Matrix

$$\begin{bmatrix} \text{city}_1 \\ \text{suburb}_1 \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_0 \\ \text{suburb}_0 \end{bmatrix} = \begin{bmatrix} 582,000 \\ 418,000 \end{bmatrix}$$

We expect the population of the city to decrease in a year

Setting up a Matrix

$$\begin{bmatrix} \text{city}_2 \\ \text{suburb}_2 \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_1 \\ \text{suburb}_1 \end{bmatrix} = \begin{bmatrix} 565,440 \\ 434,560 \end{bmatrix}$$

The next year, we expect the population of the city to *continue* to decrease

Will it decrease indefinitely?

Setting up a Matrix

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_{k-1} \\ \text{suburb}_{k-1} \end{bmatrix}$$

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This is a *linear dynamical system*

To guess what the population will look like in 20 years, we need to compute

$$\begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix}^{20} \begin{bmatrix} \text{city}_0 \\ \text{suburb}_0 \end{bmatrix}$$

demo

Markov Chains

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(familiar?)

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What's special about this matrix?

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- » Its columns sum to 1

This should make us think probability

Stochastic Matrices (familiar?)

Def. A matrix in $\mathbb{R}^{n \times n}$ is **stochastic** if its entries are nonnegative and its columns sum to 1

$$\begin{bmatrix} 0.7 & 0.1 & 0.3 \\ 0.2 & 0.8 & 0.3 \\ 0.1 & 0.1 & 0.4 \end{bmatrix}$$

Markov Chains

Def. A Markov chain is a linear dynamical system whose evolution function is given by a *stochastic* matrix

$$\vec{v}_i = A \vec{v}_{i-1}$$

Key Property of Stochastic Matrices

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Stochastic matrices *redistribute* the "stuff" in a vector

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Thm. For a stochastic matrix A and a vector $\mathbf{v}$,

$$\text{sum of entries of } \mathbf{v} = \text{sum of entries of } A\mathbf{v}$$

Key Property of Stochastic Matrices

Thm. For stochastic A and a vector $\mathbf{v}$

$$\begin{aligned}
 \vec{v} &= \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} & A &= \begin{bmatrix} \vec{a}_1 & \dots & \vec{a}_n \end{bmatrix} & \mathbf{1}^T \mathbf{v} &= \mathbf{1}^T (A\mathbf{v}) & \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} &= 1+2+3=6 \\
 & & & & \text{sum to one} & & & \\
 \mathbf{1}^T (A\vec{v}) &= \mathbf{1}^T \left(\sum_{i=1}^n v_i \vec{a}_i \right) &= \sum_{i=1}^n v_i \boxed{\mathbf{1}^T \vec{a}_i} & & & & & \\
 & & & & & & & \text{(1} \times \text{n) (n} \times \text{1)} \\
 & & & & & & & = \sum_{i=1}^n v_i = \mathbf{1}^T \vec{v}
 \end{aligned}$$

Returning to the Example

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix} \begin{bmatrix} \text{city}_{k-1} \\ \text{suburb}_{k-1} \end{bmatrix}$$

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$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix}^k \begin{bmatrix} \text{city}_0 \\ \text{suburb}_0 \end{bmatrix}$$

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What if we want to analyze a different population?

Returning to the Example

$$\begin{bmatrix} \text{city}_k \\ \text{suburb}_k \end{bmatrix} = \begin{bmatrix} 0.95 & 0.3 \\ 0.05 & 0.97 \end{bmatrix}^k \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

60% of pop. in city
40% of pop. in suburb

We should think about **how the population is distributed** instead of a particular population

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Def. A **probability vector** is a vector of nonnegative values that sum to 1. They represent:

» discrete probability distributions

» distributions of collections of things

(these are really the same thing)

Probability Vectors (Example)

The vector $\begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$ represented the distribution of the population, but we can also think of this as:

If we choose a random person from the population we'll get someone:

in the city with probability 0.6

in the suburbs with probability 0.4

The Point

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Since stochastic matrices preserve $\mathbf{1}^T \mathbf{v}$, they *transform* one distribution into another

Can we say something about how the distribution changes in the long run?

Steady-State Vectors

Steady-State Vectors

Definition. A **steady-state vector** for a stochastic matrix A is a probability vector $\mathbf{q}$ such that

$$A\mathbf{q} = \mathbf{q}$$

A steady-state vector is *not changed* by the stochastic matrix. They describe *equilibriums*

Returning to the Example

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How do we interpret a steady-state vector for our example?

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The populations in the city and the suburb stay the same over time

Returning to the Example

How do we interpret a steady-state vector for our example?

The populations in the city and the suburb stay the same over time

The same number of people are moving into and out of the city each year

Fundamental Questions

Do steady states exist?

Are they unique?

How do we find them?

Finding Steady-State Vectors

$$Aq = q$$

Let's solve this equation for q :

$$A\vec{q} = I\vec{q}$$

$$A\vec{q} - I\vec{q} = \vec{0}$$

$$(A - I)\vec{q} = \vec{0}$$

matrix eq.

How to: Steady-State Vectors

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How to: Steady-State Vectors

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Solution. Solve the equation $(A - I)\mathbf{x} = \mathbf{0}$ and find a solution whose entries sum to 1 (this will be possible given a free variable)

If there is no such solution, the system does not have a steady state

ex.

Example

$$A \vec{q} = \vec{q}$$

$$A \begin{bmatrix} 3/8 \\ 5/8 \end{bmatrix} = \begin{bmatrix} 3/8 \\ 5/8 \end{bmatrix}$$

$$\begin{bmatrix} 0.95 & \cancel{0.3} \\ 0.05 & 0.97 \end{bmatrix}$$

0.03

$$A - I = \begin{bmatrix} 0.95 & 0.03 \\ 0.05 & 0.97 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.05 & 0.03 \\ 0.05 & -0.03 \end{bmatrix} \sim$$

$$\begin{bmatrix} -5 & 3 \\ 5 & -3 \end{bmatrix} \sim \begin{bmatrix} 5 & -3 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -3/5 \\ 0 & 0 \end{bmatrix}$$

$$x_1 = 3/5 x_2$$

$$x_2 \text{ is free}$$

$$x_1 + x_2 = 1$$

$$3/5 x_2 + x_2 = 1$$

$$x_1 = 3/8$$

$$x_2 = 5/8$$

$$\Leftrightarrow 8/5 x_2 = 1$$

demo

Existence vs Convergence

If $(A - I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then the system has a stable state. This **does not** mean:

- » the stable state is unique
- » the system *converges* to this state

Convergence

Convergence

Def. For a Markov chain with stochastic matrix A , an initial state $\mathbf{v}_0$ **converges** to the state $\mathbf{v}$ if $\lim_{k \rightarrow \infty} A^k \mathbf{v}_0 = \mathbf{v}$

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Def. For a Markov chain with stochastic matrix A , an initial state $\mathbf{v}_0$ **converges** to the state $\mathbf{v}$ if $\lim_{k \rightarrow \infty} A^k \mathbf{v}_0 = \mathbf{v}$

As we repeatedly multiply $\mathbf{v}_0$ by A , we get closer and closer to $\mathbf{v}$ (in the limit)

Example of Non-Convergence

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Non-Example. I is a stochastic matrix and

$$I\mathbf{v} = \mathbf{v}$$

for any choice of $\mathbf{v}$

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So this system does not have a unique steady state

And no vectors converge to the same stable state

Regular Stochastic Matrices

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Definition. A stochastic matrix A is **regular** if A^k has all positive entries for *some nonnegative* k

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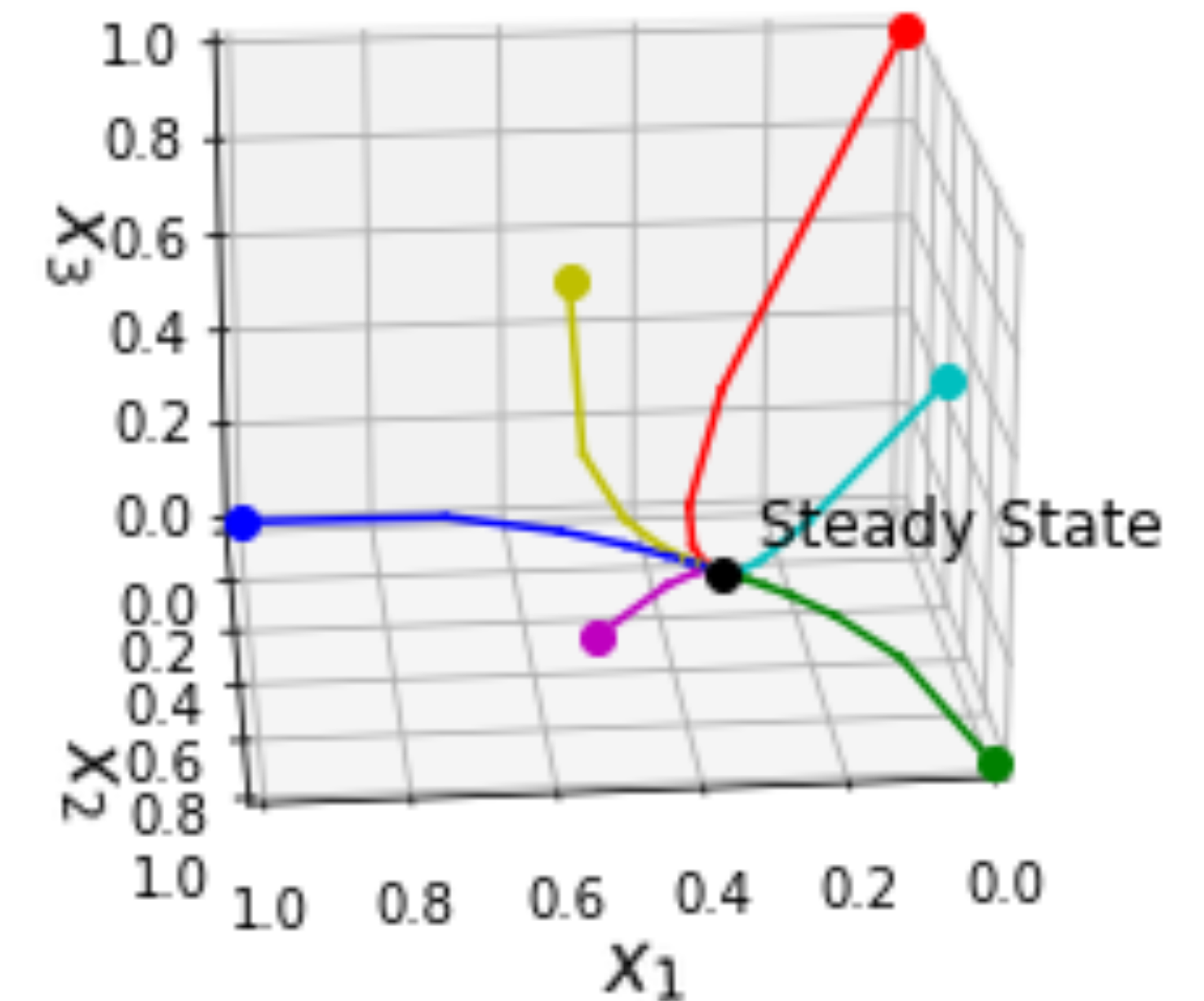
Theorem. A regular stochastic matrix P has a unique steady state, and

every probability vector
converges to it

Mixing

This process of converging to a unique steady state is called "mixing"

This theorem says, after some amount of mixing, we'll be close to the stable state **no matter where we started**



Example

$$\begin{bmatrix} 0.5 & 0.4 & 0 \\ 0.5 & 0.4 & 0.5 \\ 0 & 0.2 & 0.5 \end{bmatrix}$$

State Diagrams

Def. A **state diagram** is a directed weighted graph whose adjacency matrix is stochastic.

Ex.



Naming Convention Clash

The nodes of a state diagram are often called **states**

The vectors which are dynamically updated according to a linear dynamical system are called **state vectors**

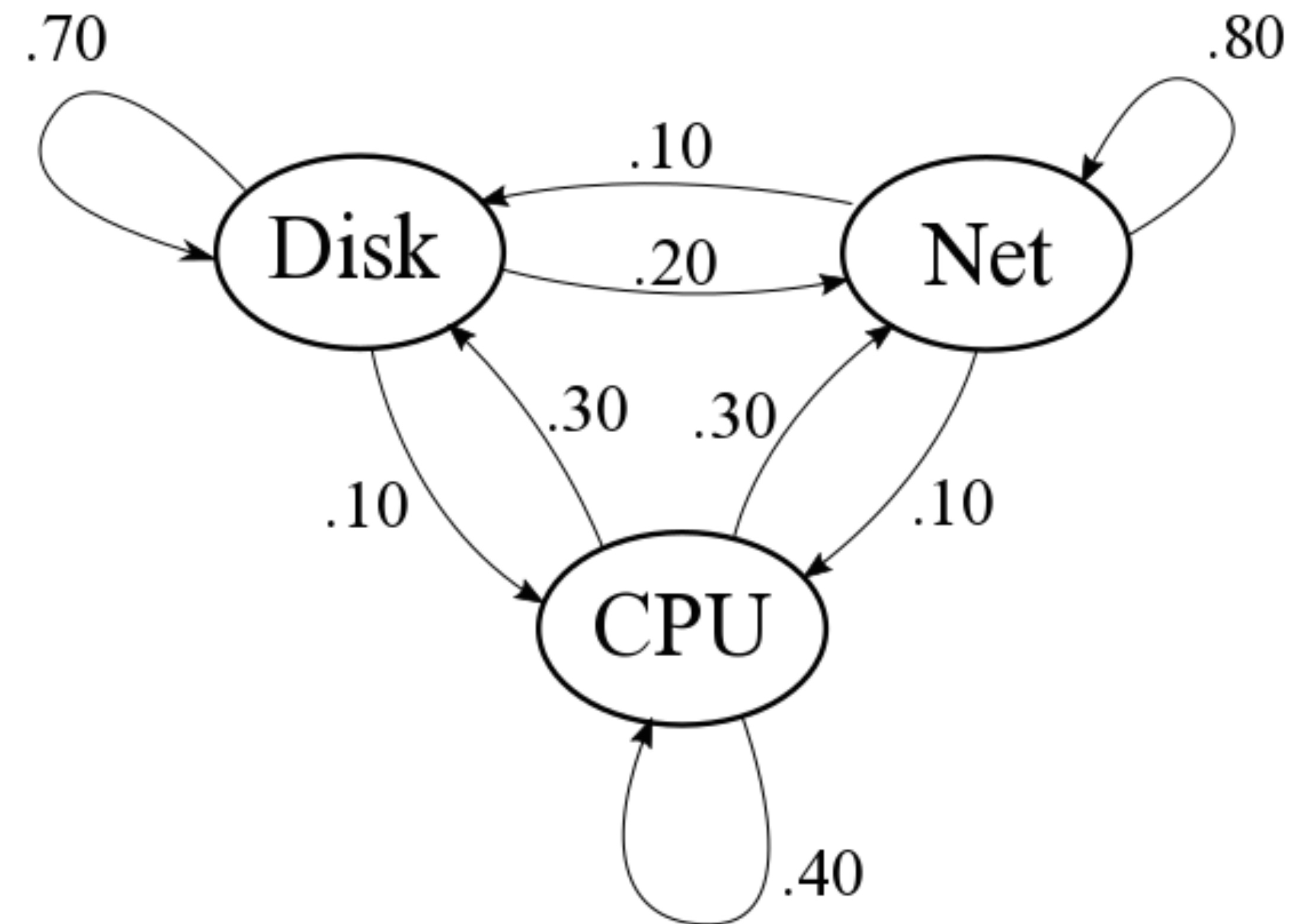
This is an unfortunate naming clash

Example: Computer System

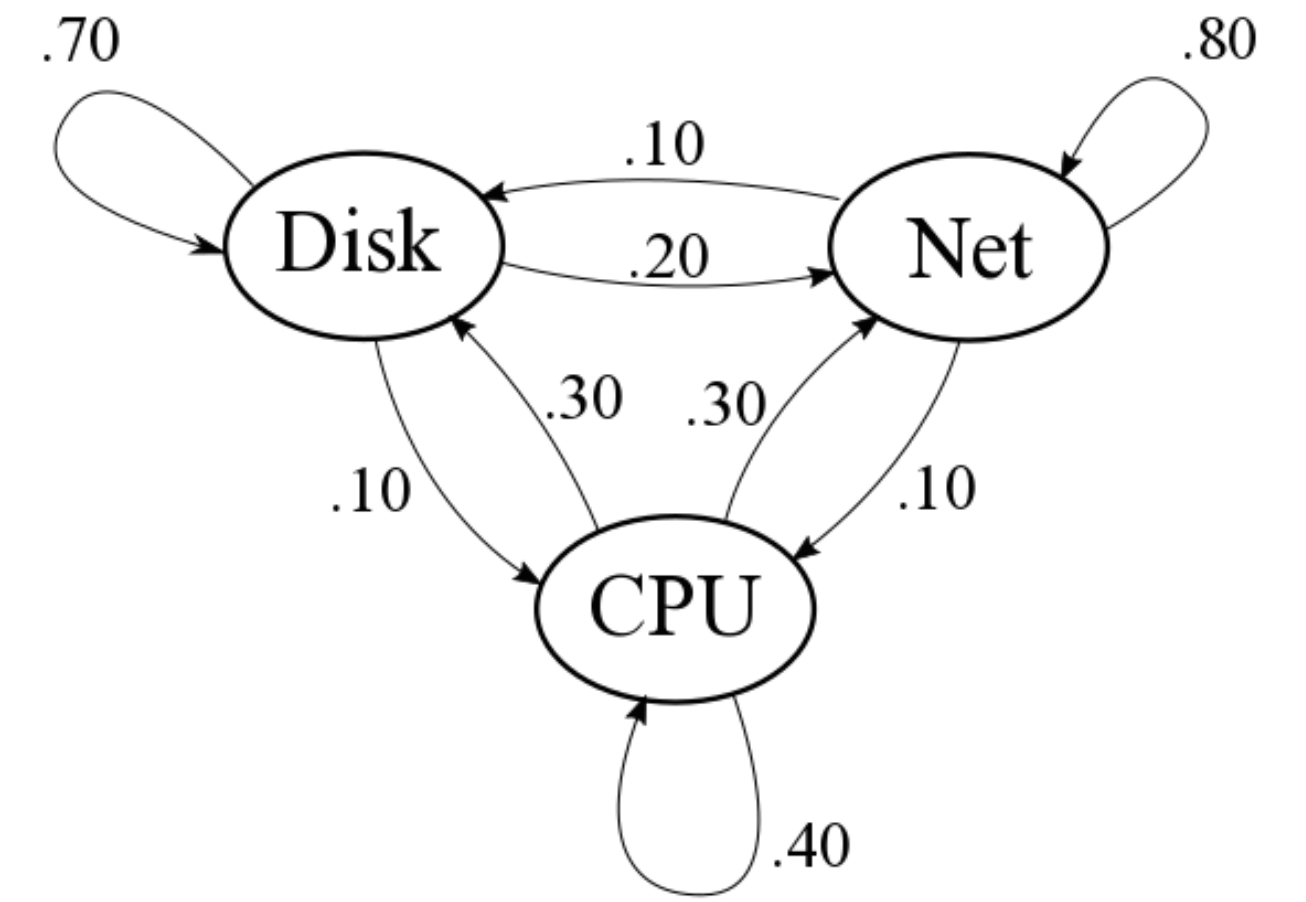
Imagine a computer system in which tasks request service from disk, network or CPU

In the long term, which device is busiest?

This is about finding a stable state



Example



Summary

Markov chains allow us to reason about dynamical systems that are dictated by some amount of randomness

Stable states represent equilibriums